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Bettina Ballin



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Experience

Freelance IT Consultant, Software & Consulting — Remote / Germany, Apr 2025 – Present

- Architecture and optimization of customized software solutions for SMEs
- Administration and hosting of IT infrastructures, including web projects
- Consulting businesses on the implementation of digital tools and technologies to enhance efficiency and security

Fraunhofer SIT, Research Associate, Secure Software Engineering — Darmstadt, Nov 2022 – Apr 2025

- Collaboration in interdisciplinary teams to develop secure software systems and prepare research reports
- Authoring scientific publications on AI and cybersecurity
- Research and development of AI-based anomaly detection systems for the German aerospace industry
- Conducting penetration tests for government agencies to identify and mitigate security vulnerabilities

Software AG, webMethods Consultant — Darmstadt, Feb 2018 – Jun 2020

- Consulting and supporting IT teams of leading German banks and insurance companies in integrating complex systems
- Implementing backend and frontend solutions using the webMethods integration platform to connect systems
- Active participation in agile projects (Scrum, Kanban) and ensuring technical implementation of requirements
- Training clients in the use of integration platforms and supporting the adoption of new technologies

Technical University of Darmstadt, Student Assistant — Darmstadt

- **Tutoring for Statistical Machine Learning**: Supporting practical sessions — May 2022 – Sep 2022
- **PTW Institute**: Implementation of energy flexibility tools in Python, development of web applications using Flask and Docker — Oct 2020 – Apr 2022
- **SECUSO Research Group**: Development of Thunderbird add-ons to protect against phishing attacks — Mar 2016 – Dec 2017

Staufenbiel Institute, Event Hostess & Promoter — Hesse, Germany, Oct 2014 – Mar 2015

- Recruitment and event organization targeting young graduates

Education

M.Sc. **Technische Universität Darmstadt** — Oct 2020 – Oct 2022

- M.Sc. Informatik, Grade: 1.3 (German scale, excellent)
- Thesis: *Assessing the Robustness of Membership Inference Attacks in Neural Networks with Label-Only Exposures*
- Focus on AI technologies. Minor: Philosophy

B.Sc. **Technische Universität Darmstadt** Oct 2013 – Dec 2017

- B.Sc. Informatik, Grade: 1.9 (German scale, very good)
- Thesis: *Toolbased Detection of Malicious Links in Online E-Mail Applications*

Abitur **Ulrich-von-Hutten-Gymnasium, Schlüchtern** Aug 2005 – Jul 2013

- Allgemeine Hochschulreife, Grade: 2.3

Certificates and Awards

- T.I.S.P. TeleTrust Information Security Professional (2023)
- Deutschlandstipendium (2020)
- webMethods 9.9 Certified ESB Developer (2018)

Skills

- **Programming Languages:** Java, Python, C/C++, C#, SQL, LaTeX
- **Web Development:** HTML, CSS, JavaScript, Angular, Flask, jQuery
- **Tools & Technologies:** Docker, Git/GitHub, Jira, Confluence, Visual Studio Code, Eclipse, Android Studio
- **Databases:** Oracle, MongoDB
- **AI & Machine Learning:** TensorFlow, PyTorch, NLP, Computer Vision, Anomaly Detection
- **IT Security:** Penetration Testing, Vulnerability Analysis, Cybersecurity
- **Operating Systems:** Linux, Windows
- **Additional Skills:** Agile Methodologies (Scrum, Kanban), Technical Documentation, Microsoft Office

Languages

- **German**: Native language
- **English**: Fluent
- **French**, **Spanish**: Intermediate
- **Chinese**: Basic proficiency

Miscellaneous

- Organizer and mentor during orientation events at TU Darmstadt (Winter semester 2016)
- Interests: Piano, logical puzzles, yoga, bouldering
- Strengths: Quick learner, passion for solving software-specific and mathematically technical problems, high resilience